Program: **Diploma in Food Processing Technology**	
Course Code: **6422C**	Course Title**: Processing of Meat and fish**
Semester**: 6**	Credits: **4**
Course Category: **Open Elective**	
Periods per week: **4(L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

1. To understand various equipments and processes involved in meat and fish processing industries.
2. To study safety and sanitation related to the meat and fish industries
3. To impart knowledge on various techniques of processing and preservation of meat and fish

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of Meat	meat and fish

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Categorize various equipment involved in production, processing and acceptance of meat and poultry.	16	Understanding
CO2	Explain the principles of various meat processing technologies, including chilling, freezing, canning, and pickling, as well as meat plant sanitation and safety regulations (MFPO) and by- product utilization in meat and poultry processing.	12	Understanding
CO3	Discuss about the care required in handling and transportation of fish at sea and on land, as well as the operation of shipboard refrigeration equipment and cold chain management	18	Understanding

CO4	Explain the processes involved in salt curing, storage, and quality assessment of fish, as well as production of cold-smoked, hot- smoked, and smoked fish using curing liquid.	14	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Categorize various equipment involved in production, processing and acceptance of meat and poultry.**		
M1.01	Discuss current level of production, consumption and export of meat products and classify the equipments	5	Understanding
M1.02	Outline the effect of feed, breed and management on production and quality	5	Understanding
M1.03	Discuss meat colour and flavours	3	Understanding
M1.04	Summarize meat microbiology and safety	3	Understanding
Contents: Machineries for processing of meat products, Slicer: Mincer: Bowl chopper Tumbler: Mixer/ Massager: Brine injector Sausage filler Current level of production, consumption and export of meat products. Effect of feed, breed and management on production and quality, meat colour and flavours, meat microbiology and safety			
CO2	**Explain the principles of various meat processing technologies, including chilling, freezing, canning, and pickling, as well as meat plant sanitation and safety regulations (MFPO) and by- product utilization in meat and poultry processing.**		

M2.01	Explain various Processing technologies for meat	3	Understanding
M2.02	Outline regulations and procedures for Meat plant sanitation and safety	4	Understanding
M2.03	Describe about by-product utilization of meat and poultry processing.	4	Understanding
	Series Test –I	1	
Contents: Various Processing technologies for meat-chilling, freezing, canning, pickiling. Regulations and procedures for Meat plant sanitation and safety-MFPO. By-product utilization of meat and poultry processing			
CO3	**Discuss about the care required in handling and transportation of fish at sea and on land, as well as the operation of shipboard refrigeration equipment and cold chain management**		
M3.01	Discuss the care in handling and transportation of fish at sea	6	Understanding
M3.02	Discuss the care in handling and transportation on land;	6	Understanding
M3.03	Identify ship board refrigeration equipments; cold chain	6	Understanding
Contents: Care in Handling and transportation of fish at sea, Handling and transportation on land, Ship board refrigeration equipments; cold chain			
CO4	**Explain the processes involved in salt curing, storage, and quality assessment of fish, as well as production of cold-smoked, hot- smoked, and smoked fish using curing liquid.**		
M4.01	Explain the process of salt curing in fish, including its storage methods and factors affecting the quality of the cured fish.	4	Understanding
M4.02	Explain the processes involved in the production of cold-smoked and hot-smoked fish.	5	Understanding
M4.03	Explain the process of describing smoked fish using curing liquid.	4	Understanding
	Series Test–II	1	
Contents: Salt curing of fish, storage, quality, Production of cold smoke fish; Hot smoked fish, smoked fish using curing liquid.			

Text / Reference

T/R	Book Title/Author
R1	Joseph Kerry, John Kerry and David Ledwood. ―Meat Processing , Woodhead Publishing Limited, England (CRC Press), 2002.
R2	Mead, G. Poultry Meat Processing and Quality , Woodhead Publishing, England, 2004.
R3	Gerasimov, G.V. and Antonova, MT. "Techno-Chemical Control of fish Processing Industry". Amerind Publishing Co. Pvt. Ltd., 1979.
R4	Borgess, G.H.O., Cutting, C.L., Lovern, J.A. and Waterman, U. "Fish Handling and Processing". Chemical Publishing Co., 1967
R5	Fish & Fisheries of India; Jhingram VG; 1983, Hindustan Pub Corp
R6	Fish as Food, Vol. I-IV; George Borgstrom, Academic Press 5. Fish Processing Technology , Rogestein&Rogestein

Online Resources

Sl.No	Website Link
1	https://au.teysgroup.com/food-info/what-is-cured-meat/
2	https://www.sciencedirect.com/topics/food-science/smoked-meat
3	https://seafoodhealthfacts.org/safety/seafood-handling-and-storage/
4	https://www.fao.org/faolex/results/details/en/c/LEX-FAOC152883/